



RE:CREON

Player's Manual

ANACREON: RECONSTRUCTION 4021

EDITION 0.1.0 — 2026-10-01

A FAITHFUL RECREATION IN PYTHON

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1 About This Manual

The empire is not a battle. It is a budget that occasionally fights. – *Imperial Staff primer*, author unknown

Re:creon is a faithful recreation, in Python, of **Anacreon: Reconstruction 4021** – a turn-based 4X space strategy game first published in 1990 and released as Turbo Pascal source code in 2004. Every number, table and rule in the game you are about to play was transcribed from that source rather than reimagined, which makes the original’s quirks as much a part of this game as its good ideas.

This manual replaces the printed booklet that came in Anacreon’s box. The original shipped no tutorial and its in-game help was a stub; the paper manual was the only way to learn the game. This version covers the same ground as that manual’s index – combat, construction, defenses, fleet orders, materials, ships, technology, world classes – and adds what the old one took for granted: how to actually install and start the game, and how this recreation’s screens work.

1.1 What kind of game this is

You run an empire. Time moves one **year per turn**, and a turn is divided into orders you give – to worlds, fleets, construction, diplomacy – and a year of consequences you watch unfold in the news window. Each of up to eight empires is run by a player or by one of several computer personalities; in the default mode you play one empire and the others act after you.

The levers are the classic 4X ones, but filtered through a 1990 design sensibility that rewards reading tables:

- **Explore.** Your sensors see only what your fleets and worlds have scouted. Starbases slowly map their neighborhoods; probes fly ahead of you.
- **Expand.** Worlds are claimed, designated for a job, and built up with industry and construction.
- **Exploit.** Every ship, base and gate has a raw-material bill – chemicals, metal, supplies and trillum – paid out of a planetary economy that can starve, riot or get addicted to ambrosia.
- **Exterminate.** Combat is resolved shell by shell, from deep space down to the ground, with a published table of what destroys what.

1.2 Conventions used here

- Key names appear as `F3`, `Enter`, `Esc`. Combinations are written out (`Ctrl+C`).
- Menu choices are written `Menu ▶ Item`, e.g. `Game ▶ Next turn`. Each menu item also has a one-letter accelerator shown in its label; the letter is not always the first one (the original chose `caNcel orders` and `auTo attack`).
- Coordinates are always **relative to your capital**, which reads as `0,0`. North is `+Y`. Sector means the same thing as star system here: one cell of the square grid the map shows.
- Chapter 8 lists every command, key and status window in one place; the reference appendix (last chapter) reproduces the game’s actual data tables, so a number quoted here and a number the simulation uses are never two different numbers.

I 1.3 Where the name comes from

Anacreon was itself named after the ancient Greek poet. *Reconstruction 4021* was the subtitle of the 2004 release – the year, in the game's own fiction, that the old empire fell apart and the reconstruction began. Re:creon keeps both halves: it is the original game, reconstructed once more.

The rest of this manual assumes you have never played the original. If you have: nothing here will surprise you, and that is the point.

2 Getting Started

The first order of government is the map. The second is the budget. The third, unfortunately, is also the budget. — *First manual of the Reconstruction fleet*

2.1 Installing and running

Re:creon needs Python 3.12 or later and [uv](#) to manage dependencies. From a clone of the repository:

```
uv sync          # install dependencies (first time only)
uv run recreon   # launch the game
```

The game opens full-screen in your terminal – a Textual interface that replaces the original's DOS screens one for one. Any terminal of at least 80x24 works; a truecolor terminal looks best.

Useful command-line options:

Option	Effect
<code>--scenario FILE.SCN</code>	Skip the scenario picker and start this game
<code>--scenario-dir DIR</code>	Where the picker looks for scenarios
<code>--name NAME</code>	Name your empire instead of choosing interactively
<code>--save-dir DIR</code>	Where saved games are written and read
<code>--no-ui</code>	Advance exactly one turn headlessly and print the result
<code>--version</code>	Print the version and exit

2.2 The prologue

What you see first is the **prologue**: the menu that lives outside any game. Its rows, in order:

- **Begin...** – enter the game that is loaded, if any.
- **New game** – pick a scenario, page through its introduction, choose how many empires are run by humans, and name them. Computer empires get their names from the original's 59-name table; you can accept a random name or type your own.
- **Load game** / **Save game** – saved games are JSON files in the save directory. (The original's save format is a raw dump of in-memory records; nothing in it could survive the translation to Python, so the saves here are readable, versioned files instead. A DOS-era save game cannot be loaded.)
- **Quit** – leaves the program, asking first if the game has changed since the last save.
- **Time limit** – seconds of thinking time added to each player's turn per year, for anyone who wants a clock.
- **New player empire** / **Delete player empire** – adjust how many empires humans run.
- **Auto backup** – whether the game autosaves after every turn. It is ON by default, and the autosave is how the original worked too: a turn that has ended is a turn that is already saved.
- **Pause** – whether a year's news can pause for reading.
- **Sequential play** – ON rotates through empires in a fixed order (Empire 1 through 8, then the year turns). OFF is the play-by-mail mode: each human takes their turn from the prologue, hands the save file on, and the year only turns when the last human has moved. For a solo game against the computer, leave it ON.

2.3 The main screen

Once a game starts you are looking at the **strategic map**: a square grid of sectors, each a star system. Worlds, bases, gates, fleets and phenomena are single-character glyphs on a grid – exactly as the original drew them, because the original fit the whole game into 80x25.

- The cursor is your point of attention. Move it with the arrow keys or **h j k l**.
- Coordinates read **relative to your capital**, which sits at **0,0**; +Y is north. Another empire's capital has its own **0,0** from its own point of view, never yours.
- The menu bar carries the seven command menus – *Info, Game, Empire, Worlds, Fleet, Build, Ministry of War* – documented in full in chapter 8. Press **m** or **F2** to reach it.
- Direct keys skip the menu for the most-used screens: **f** fleets, **b** construction, **w** warp links, **d** defenses, **x** self-destruct, **z** close-up of the cursor's object, **i** production report for it.
- The **F** keys raise the status windows: **F3 / F4** world and military status, **F5 / F6** fleets, **F7** news, **F8** empires, **F9** names and records, **F10** the map. **F1** is help.
- **n** ends your turn and runs the year.

2.4 What you can see: the fog of war

Knowledge comes in three grades, and they matter everywhere:

1. **Unknown** – you have no idea the object exists.
2. **Known** – you know something is there; not what it is. The map shows known objects.
3. **Scouted** – you can see its contents. Detail panels, the status tables, and targeting all require scouting, not merely knowing.

Each turn, what your empire *observed* is recomputed from scratch: fleets scout their own sector and report, and every world scouts what it can see around itself. An empire that withdraws its fleets genuinely loses sight of what it was watching. Objects far from anything you own stay unknown until a **starbase scan** turns up one – and a base's scan is a coin flip per object per year, so a new base maps its neighborhood over several years. **Probes** (*Fleet ► Probe*, up to ten per empire) fly ahead and carry word of what they find.

2.5 A first turn

When a scenario starts, do this:

1. Press **z** on your capital's sector and read the close-up: class, population, technology, defenses, what is under construction.
2. Press **i** for the production report and see what your worlds are making. Ships are not built at sites – a world with shipyards turns raw materials into hulls every year on its own. Sites (**b**) build the eight big structures: SRM fields, bases, fortresses, complexes, outposts, gates, links and disrupters.
3. Press **f** to see your starting fleets; select one, and give it a destination with *Fleet ► Change destination*. Explore something near your borders, not far from home: standard fleets cross one sector per year, and fuel is finite.
4. Press **n**, and read next year's news with **F7**.

That loop – inspect, order, advance, read – is the whole game. The following chapters go into what each part is actually deciding for you.

3 Worlds and Their Economies

A world is worth exactly what its factories say it is worth, and not one headline more. —
Ledger of the Fourth Reconstruction

Every planet is a small economy with a population, a technology level, and a stack of industries that turn raw materials into the things the empire needs. Worlds are where the game is actually won; fleets only spend what the worlds produce.

3.1 The anatomy of a world

A world's close-up ( on its sector) shows its:

- **Class** – what the world physically is: Paradise, Ice world, Gas Giant, Ancient ruins, and the rest of the twenty-two classes. The class is permanent except by terraforming (see below), and it sets the world's trillum reserve, how much its terrain helps ground defense, and the percentage of each industry that runs effectively there. The class table in the appendix lists all of them; a Paradise world is worth more than a Barren one at the same population, and the map glyph tells you which is which at a glance.
- **Designation** – the job you assign: agricultural world, metal mine, shipyard base, university, capital, and so on. Designations require a minimum technology, set the world's principal industry, and change how its leftover industry distributes. You change them with *Worlds ► Designate*.
- **Population**, counted in units of 100 million. Population grows toward the maximum its technology supports; the base figure for a world at 50% efficiency runs from 3 (a hundred million, pre-tech) to 3,000 (three hundred billion, gate technology).
- **Technology level**, 0 to 10, from pre-tech to gate. It governs what the world's industry can produce at all, and it can outrun the empire's own level – which matters for research, not just production.
- **Efficiency**, 0..100: how well the population actually works, worn down by famine, ambrosia withdrawal, occupation and shock, and rebuilt in calmer years.
- **Revolution index**, 0..100: unrest. Unrest grows with distance from the capital, with the empire's overall mood, with starvation and with garrisons on quiet worlds. Above 30 the world mutters, above 43 it marches, above 66 it rioted last week; above 75 it may rebel outright. A rebellion either gets crushed by the local troops and ninjas or succeeds, and the world goes independent. The capital never rebels. You can see the mood building in the news long before the map changes color.
- **Hold cargo**: troops, ninjas, and stocks of the four raw materials sitting in the world's warehouses, plus its fixed defenses spread across the orbital shells.

3.2 Production: where everything comes from

Each year, a world's **industrial potential** – driven by population, technology, class and efficiency – is distributed among the nine industries:

Industry	Makes
food factories	supplies
chemical plants	chemicals
metal mines	metal
trillum mines	trillum
bio-tech labs	ambrosia (and ninjas, with the right class)
ship yards	fighters and transports, mostly
jumpship yards	jumpships, hunter-killers, jumptransports
starship yards	starships and penetrators
transport yards	transports and fighters

Yards consume raw materials (the cost tables are in the appendix) and spit out hulls every year without being told; the split between what a world keeps as raw material and what it converts is set by its designation and by the **ISSP** slider.

ISSP (*Worlds* ► *ISSP*) is four sliders – one for each raw material: chemicals, metal, supplies, trillum – each set 0 to 10, with 5 meaning the world produces exactly what it needs of that material. Set a world's supplies slider to 10 and it pours industry into food factories to feed its neighbours; set it to 1 and it starves politely and lets its chemical plants eat the difference, hoping someone ships. The budget is shared, so every slider raised takes industry from somewhere else.

Raw output is a forecast (*Worlds* ► *Production*), not a promise: an exhausted trillum reserve, or a world that cannot feed itself, will produce less than the screen projects.

3.3 Feeding everyone

Supplies are the economy's heartbeat. Every billion people consume 25 megatons of supplies a year. A world that cannot produce them from its own food factories is **not self-sufficient**, and must be shipped food by fleet – run the convoy aground, and the news starts reporting starvation. Famine kills population, drops efficiency and pushes the revolution index up; in the old DOS manual's phrase, a blockaded world is a loyal world right up until it isn't.

3.4 Ambrosia

Bio-tech technology (level 7) unlocks ambrosia, and ambrosia is a trap with a payoff attached. Worlds with bio-tech labs produce it; at 11.5 kilotons per billion people a year it holds off addiction, each year a world's population has a 25% chance of becoming dependent. An addicted world kept off its drug suffers mass casualties – and the deaths drag efficiency and the revolution index down into the basement. Delivered on time, though, ambrosia makes the population work harder: an addicted-and-satisfied world runs its industry at 145%. Many late-game empires are, functionally, drug dealers with a navy. Independent worlds drift toward addiction anyway; liberating a world does not liberate its habits.

3.5 Special worlds

- **Capital** (*Worlds* ► *Designate* to move it, if you can stomach the staff's objections): +12% annual research chance, immune to rebellion, and in some scenarios the loss of it loses the empire.
- **University worlds** (designatable at jump tech) are the engine of research – see chapter 4.
- **Ancient ruins** worlds from the old empire yield a small research bonus and, according to scenario authors, occasionally something worse.
- **Hostile life** worlds have natives: a strong garrison deters them, and under some empires they can be enlisted as ninja legions instead of fighting them.
- **Terraforming** worlds are mid-transformation – see below.

You cannot designate every world every job: universities need jump technology, ambrosia worlds need bio-tech, ninja worlds need starship tech, and the gate-tier variants of every base type need gate tech. The appendix table gives the minimums.

3.6 Terraforming

Worlds ► Terraform (warp technology, and a world already paid for in metal and patience) rewrites a world's class. The target class is drawn from a short list each class may become – a Desert can turn greener, an Ice world can be melted – and harder targets terraform more slowly: Paradise, the best land there is, completes on about one annual roll in ten, and every attempt carries its own standing chance of catastrophic failure, which leaves the world something worse than it started. While the job runs the world is a Terraforming world: half its former industry and a fraction of its efficiency. Finish it and you own the empire's best kind of real estate: a new Paradise. Fail to protect it and the enemy inherits it.

3.7 Designing for the long run

Three habits separate a Reconstruction that outlives its first century from one that dissolves into the news archive:

1. **Match designations to classes.** A trillum mine on a Desert runs at 190% strength; the same mine on a Poisonous world runs at 80%. The class table in the appendix says exactly where each industry earns its keep.
2. **Watch the revolution index like fuel.** Garrisons put down revolts, but quiet worlds resently garrisoned push unrest the other way; military index is a dial, not a setting.
3. **Feed the labs, then the yards.** Production you cannot scout with is production the enemy will take delivery of.



4 Technology and Research

No one steals a civilization's tools from an armory that has nothing left to lose. – *Guardian proverb, date of composition disputed*

4.1 Eleven levels

Technology runs from **0, pre-tech**, through primitive, pre-atomic, atomic, pre-warp, warp, jump, bio-tech, starship and pre-gate, to **10, gate** – the drive technology that scattered the old empire and the ceiling of what anyone here can reach again.

Each level unlocks its kit: atomic tech opens trillum mining and ground-defense missiles, pre-warp gives you fighters and command bases, warp gives transports, jump gives the jumpships and the bases that carry them, bio-tech opens hunter-killers, penetrators, ambrosia and outposts, starship tech brings the capital ships, LAMs and ninjas, and gate tech – fifteen years a site – brings stargate links and disrupters that rewrite the geometry of the whole map. The appendix's technology table lists exactly what each level first makes possible.

Every **world** carries its own technology level as well, and the two advance together only through the empire's research effort.

4.2 How research actually happens

Once per year, each active empire rolls for discovery. The chance is built from the empire's **labs**, of which there are four kinds:

Lab	Annual chance
Capital world	12%
University world (at the empire's level)	15%
Any world that has out-researched the empire	15%
Ruins world	5%
University standing on ruins	17%

Starbases count too, with a shorter list and no ruins bonus. Only the twenty best labs count at all.

The roll is **breadth before depth**. An empire that is missing any single technology of its current tier is spending its research on that hole – you cannot skip a level by getting lucky – and only an empire whose tier is complete can advance to the next one at all. The last technology of a tier is the lock on the door. When the empire does advance, its capital and every university and lab world come up one level with it, which is how a research empire snowballs instead of leaving its home worlds behind.

The third row of the table is the interesting one: **conquest is research**. A world that already knows more than your empire does is a working lab whatever its designation, and seizing one makes every future roll better. Losing a frontier world that out-grew your tier stings twice.

One honesty note, because the game keeps the original's arithmetic: the annual research chance is accumulated in a small integer that can overflow. Up to a point more labs mean faster research; past that point, adding another university can make the counter wrap and the empire research *worse*. The twenty-lab cap exists in the game as a mercy the DOS build could not offer.

4.3 Independent worlds

Worlds do not sit still either, but they follow the capital rather than the empire's books: any world *below* its capital's level has a 16% annual chance to climb one, and a world that has outrun the capital regresses slowly – technology out there is sustained by the centre, not held permanently. Independent worlds have no capital and advance on a bare 1-in-50 annual roll, which is still not nothing: a backwater you leave alone for two hundred years may be embarrassing to conquer by then. This is also why “wait and see” is not free: the galaxy gets richer while you hold still.

4.4 Trading technology

Empire ► *Trade technology* hands one of your technologies to another empire – free, no request in return, and no check whether they already have it: two empires at the same level are each always offering the other their entire tier. Gift technology to a small neighbor to keep them ahead of a large one, or watch what the pirates do with a jump drive. It is also the only way to give someone something *after* you conquer their lab world – a peace offering the recipient can read as exactly what it is.

4.5 What tech gates

Technology is checked in four places, all of them annoying at the worst moment:

- **Designations** – universities need jump tech to found, ninja worlds need starship, the gate-tier bases need gate tech.
- **Industry** – you cannot run what you cannot name: bio-tech labs need level 7, shipyards at least level 4.
- **Population** – some classes cannot support life at all below a given level; Gas Giants and Oceans wait for pre-warp.
- **Ships in space** – a world's scouts only credit it with ships its own technology could have built, so a level-4 empire reporting a level-8 starship is being honest about having *seen* something, not about what it was.



5 Construction

■ A gate never built is a war never won cheaply. – *Ministry of War memorandum, year 4019*

Shipyards make ships on their own, every year, from raw materials – that is production, covered in chapter 3. **Construction** here means the eight big installations that take a site, years, and a steady delivery line of cargo:

Structure	Years	What it is for
SRM field	2	Mines the sector; damages fleets that jump through it
Command base	6	A mobile starbase: scouting, scanning, fleet support, jump base
Fortress	12	The immovable one: the strongest fixed defense in the game
Industrial complex	10	Links the surrounding mine worlds and concentrates their output
Outpost	3	A cheap sensor and a flag planted in someone else’s back yard
Stargate	15	The jump corridor itself – the reason level-10 tech ends wars
Warp link	5	A one-way door between exactly two friendly bases
Jumpspace disrupter	8	Denies enemy jump movement around itself

A site consumes its materials **per year while it stands** (the table is in the appendix), and an unfinished site is a target: enemies can destroy one as easily as a world, and *Build ► Abort construction* tears yours down if the money stops making sense. The universe holds fifty sites at a time; when *New construction site* says there is no room, somebody else’s project is using it.

5.1 Bases

Command bases and fortresses carry destinations like fleets and crawl one sector a year, burning trillum whether blocked or not; all other base types are fixed once built. Bases of any kind:

- scout what is around them and scan for what is not, which is the only way to learn about objects your fleets never visit;
- act as the jump base for fleets sitting on them;
- refuel fleets;
- cannot enter an occupied sector at all – and a base being shoved sideways by its own pathing can actually be pushed off the playable grid, because the original checked its detours nowhere. Watch bases that hug the edges.

An **industrial complex** is a different animal: it is the grid connector between the eight neighbouring sectors. Each year it pulls raw material from friendly mine worlds that hold a working reserve, and pushes its own overflow back out to the same mines when its hold is full. Clustering chemical plants,

metal mines and trillum mines around a complex is how late-game empires pay for gate construction without bankrupting a sector.

5.2 Gates, links and disrupters

A **stargate** sends jump-capable fleets – jump fleets and hunter-killer forces, not standard warp traffic – between any two gates tuned to the same **frequency**, a number 0 to 9999. Frequencies are the diplomacy of movement: *Empire* ► *Link frequencies* retunes a gate, so a captured enemy gate can be folded into your network – or a retune can strand every fleet that was counting on the old pairing.

A **warp link** is a gate built as a pair: it connects your base to exactly one far end, permanently, and needs none of the frequency diplomacy a stargate does. Five years instead of fifteen; a corridor you can widen only by building more of them.

A **jumpspace disrupter** makes the space around it hostile to jump drives: enemy jump fleets cannot make jump moves through its field, while *your own* warp-driven fleets riding inside a friendly disrupter's field move at jump speed – ten sectors a year on hulls that normally crawl one. A disrupter ringed around a choke point is the strongest defensive structure in the game and the most expensive thing to fight through.

5.3 Fortresses, gates and the two rules that surprise people

A **fortress** multiplies the value of everything a defended sector is worth – the combat table's base adjustment makes a fortress sector worth two and a half times an open one – and at twelve years' construction, it is a statement about where you intend to stand and not move. It also does something no manual written before 1990 mentioned politely: **a fleet that departs from under a fortress is catapulted four sectors along its way** before it even starts. Fortresses are not defense. They are short-range logistics with a gun on top.

A fleet entering a stargate arrives the next year, whatever the distance. One sector away or across the map – gates move jump traffic at a flat one year per jump, and only the two ends need their frequencies in agreement. Ten-sector jumps from jump bases pale next to this, which is why gate construction (fifteen years) decides wars decades before anyone fires in the sector they connect.

SRM fields (space minefields) damage anything that *jumps* through the sector – hunter-killers, jumpships, jumptransports – and do nothing to warp traffic, which makes them chokepoint tools along gate lanes rather than area denial in general. They are also mortal: *Fleet* ► *SRM sweep* clears a field the slow way, one careful pass at a time, and an unescorted sweeper in enemy space is a letter the enemy will never forward.

>> 6 Fleets and Movement

Amateurs discuss tactics. Professionals discuss fuel. – *inscription, unofficial, on the wall of the Logistics Directorate*

6.1 What a fleet is

A fleet is a named pile of ships and cargo sitting in one sector – yours or nobody's – with a destination, a fuel load, and standing orders. *Fleet* ► *Deploy* creates one out of the ships and cargo at any object you select: a world, a base, or another fleet. Each empire can field thirty fleets; the map can hold two hundred and forty between all of them.

Fleets are not uniform. A fleet's **type** is worked out from what it *lacks* – one transport hides inside a battle fleet and drags the whole thing down – and the type decides the speed:

Fleet type	Composition	Sectors per year
Standard fleet	contains transports, fighters or starships	1
Advanced-warp fleet	penetrators and jumpships, nothing slower	2
Penetrator force	only penetrators (with escorts of hunter-killers)	2
Jump fleet	jumpships and jumptransports (and hunter-killers), no sublight hulls	10
Hunter-killer force	hunter-killers alone	10

Movement is measured with a **Chebyshev metric**: the cost of a move is the larger of the horizontal and vertical distance, so a diagonal sector is exactly as cheap as an orthogonal one and fleets travel in straight lines, diagonally included. Your coordinates are relative to your capital, +Y north; another empire's coordinates are relative to theirs, and typing one into your destination prompt gets you what you deserve.

6.2 Fuel: the trillum treadmill

Every ship's hold is part fuel tank: each type has a capacity and a consumption per sector, and **fuel is trillum** – the only fuel there is. Fleet movement burns it; standing in space burns a tenth of the same per-year rate whether the fleet moves or not. A fleet that runs dry stops dead and reports its status as *out of trillum*, and the only way out of that is *Fleet* ► *Refuel* on a friendly world or base with reserves.

The numbers have teeth. A hunter-killer sips fuel by sector and carries a deep tank – that is what raiders are made of. A starship drinks 1.4 times a transport's rate per sector and moves at half the speed of anything crossing its own space, which is why capital fleets live off base chains. And raw trillum as cargo is expensive to ship: 100 kilotons a transport.

Refuel early, refuel where the reserves are: every world's class sets how much trillum it started with (a Barren world's 1,200 beats your Paradise's 600), reserves do not grow back fast, and a fuel picture is a strategic map in its own right. The scouts' reports of enemy reserves, remember, are estimates – the status windows show another empire's holdings in a rough notation (see chapter 8), and your own in exact hundreds.

6.3 Cargo and loading

Ships that carry: transports (5 legions, 3 megatons of chemicals, 2 of supplies), jumptransports (a fifth of a transport), and cargo itself. *Fleet ► Transfer* moves ships or cargo between two fleets in the same sector. When you deploy an attack, the grouping screen auto-loads troops into empty transport slots in the groups – and beware the original's temper: changing a slot's ship type, or its count, spills that group's whole cargo back into the common hold, soldiers included. Load your groups last, or load them and then leave them alone.

6.4 Standing orders

Fleet ► Orders compiles a program for a fleet – the original's most ambitious interface, and still the fastest way to run a police force:

DEST <coordinates or name>	set destination
TRANS <resource> <amount>	transfer cargo at arrival
SRMS	sweep minefields
WAIT <years>	sit
REPE	repeat the order list
ABORT	return to base
DEST <place> / TRANS / ...	chain as many lines as you like

Compiled orders move fleets and dump cargo automatically each year without touching anything; a supply fleet on a repeat loop `DEST convoy route / TRANS sup 2000 / WAIT 1 / REPE` is a convoy that never needs a signal officer again. *caNcel orders* wipes the program. Orders break silently on arrival if the target has vanished – read the news, it will tell you.

6.5 Probes

Fleet ► Probe pokes a destination you already know with a sensor pack: up to ten per empire, each launched probe lands its news the following turn and is ready again. Probes are cheap knowledge about specific places; starbase scans (chapter 2) are patient knowledge about areas; and neither is a substitute for a fleet whose orders say *DEST and find out*.

6.6 Hazards of the deep

- **Nebulae** blind: any nebula hides what sits in it from remote sensors; a **dark** nebula additionally stops a scout's line of sight, so worlds behind one stay unknown; a **dense** nebula is a wall – warp movement into it simply fails, and a jump that ends in one makes the news instead of the destination.
- **Black holes, pulsars, wormholes** damage or vanish anything passing through; wormholes also move things, to somewhere.
- **Minefields** (chapter 5) hit jump traffic only.
- **Enemy disrupters** stop jump movement outright inside their field – friendly ones, as noted, speed *warp* traffic to jump pace instead.



7 Combat and Conflict

The shells are the same five in every system in the galaxy. Only the arithmetic changes. –
Gunnery handbook of the Reconstruction, first edition

Combat happens when your fleet and someone else's object share a sector. Two commands reach it from the map, both under *Ministry of War*:

- **Attack** opens the interactive battle: you command each group round by round.
- **auTo attack** hands the whole fight to the same tactical routine the computer empires use and shows you the result. If you have ever wondered what the AI thinks of your formation, this is the button that says so.

Defensively, there is no button: enemy computer empires attack you the moment they weigh the odds as favourable, and the fight simply appears in the news. Position is diplomacy.

7.1 Who can be fought

Attacks target enemy fleets in the sector, then enemy objects. One rule surprises people the first time: **a fleet in orbit screens the world beneath it**. If hostile fleets are present, the world is not offered as a target at all – you fight the navy first, and only then the planet. The same is true in reverse when you are defending: one surviving corvette in low orbit keeps your capital legally untouchable, which is occasionally worth more than the corvette.

7.2 The five shells

Everything in a fight stands at one of five altitudes, and combat resolves from the outside in, one round at a time:

1. **Deep space** (DpSpc) – where fleets arrive.
2. **High orbit**
3. **Orbit** – where starbases and defense satellites live.
4. **Sub-orbit** – the catch-all layer; the defense grid puts any remainder here.
5. **Ground** – where populations die.

Groups start in deep space with no targets. Ships at the same shell as a target shoot at it; shells that are uncontested let the fight descend toward the surface. This means combat is mostly a race and an alignment problem: the attacker must *close*, and the defender survives by making every shell above the ground cost more than it is worth.

7.3 The arithmetic

Each round, every group fires at its target. The damage a weapon does is looked up in the combat table – attacker type against defender type – and modified by the attacker-vs-defender technology table, the defender's terrain (ground fighting only), and any base in the sector (command bases 150%, fortresses 250%, outposts 125%). Points accumulate against each defender; **100 points destroy one unit**, 200 destroy two, and so on. The full tables are in the appendix; the short version is that a fighter squadron does 15 points of damage to a hunter-killer where a starship does 250 to the same target – but the fighter costs a twentieth the metal and is worth five points to your bookkeeping against a starship's two hundred. Cheap hulls swarm; expensive hulls win the rounds nobody can afford to repeat; transports are cargo that happens to be on fire.

A world under attack answers with its **ground-defense missiles** – a salvo sized by the world's technology (up to 2,759 at gate tech), of which the attackers shoot a fixed number down on approach per ten ships depending on hull type. Fighters catch nothing; starships catch twenty apiece.

The **defense grid** (**d** on any world you own) is where you decide what that fight looks like before it starts: percentages of each of the four fixed defenses – LAMs, defense satellites, GDMs, ion cannons – distributed across the five shells so that each type's five numbers total 100. The grid normalizes whatever you type, and **sub-orbit is where every rounding error and impossible request ends up**, which is worth knowing the first time your even 100/100 split comes back 33/33/33 with a stray 1 parked in sub-orbit.

7.4 Closing: the interactive battle

The attack screen lists your groups – up to nine per attack – and the enemy roster. Groups do nothing unless you tell them to:

- **m** manoeuvres a group: advance a shell, retreat a shell, or hold station.
- **t** gives a group its target.
- **e** fights the round.
- **d** shows the damage detail from last round, and **r** extracts everyone.

A player who only ever presses **e** will sit in a perfect battle in which nobody moves, nobody aims and nobody fires, forever. Closing and aiming are the entire point of the screen – the original's author put it in the manual's place where the help file could not: the computer aims for you only when you let it.

When you load transports for an assault the grouping screen auto-loads troops for you, with the two caveats from chapter 6: the auto-loader prefers men over ninjas, and changing a group's ships spills its cargo back to the hold.

7.5 After the shooting

Resolution cascades down: fleet wins the shells and the world is next; a beaten world's survival turn comes its own arithmetic of population, morale, the power of its remaining weapons and the size of the occupying force – worlds that lose their navy are liable to sue for terms rather than die on the ground, and the technology band a world sits in changes how often terms are offered. A conquest puts the world in your empire: population and industry change hands intact, morale shock and revolution index arrive together, and the news runs a headline either way. Scenarios with written background text give conquered worlds their own epitaphs.

Losses are tallied where you can read them – the news headlines carry the casualty figures, and the empire status and records windows (**F8** , **F9**) report what survived – and destroyed construction sites and stargates are announced as destroyed whether or not the wreckage agrees.

7.6 The tools you use to say no

- **LAMs** (*Ministry of War* ► *Launch LAMs*): light attack missiles, stockpiled as a base's fixed defense and fired without moving a fleet or risking a fight. Against an enemy fleet the salvo spreads by what each hull is worth, concentrating on the ships actually worth killing – a LAM barrage can clear a sector's navy before the invasion starts; against a world it strips fixed defenses evenly. The missiles are spent whether they hit or not, the target cannot intercept, and the news will be very clear about who fired. It is the cheapest violence in the game and the least deniable.
- **Self-destruct** (**x** / *Worlds* ► *Self-destruct*): what you do to your own world when it is clearly about to change hands. Industries die, population dies, the enemy inherits a rock.

- **Minefields, disrupters, fortresses:** chapter 5. A war won in the construction menu is a war not fought.



8 Command Reference

The map is the argument. Everything else is paperwork. – *marginalia, Imperial staff college exercise files*

This chapter is the whole interface in one place: every key, every menu item, every window. Commands are entered by picking them off the menu and answering prompts – the shipped original never had a typed-command parser, and this recreation does not add one.

8.1 Function keys

Key	Opens
F1	Help window – the original’s topic index
F2	Menu bar (same as <code>m</code>)
F3	World status window
F4	Military status window
F5	Fleet status window
F6	Fleet contents window
F7	News window
F8	Empire status window
F9	Names and records window
F10	Map (closes whatever window is open)

F3 and F4 are one window: the world status table stacked over the military status table, both listing the same worlds in the same order, so the two halves line up row for row. F5 and F6 are the same trick with fleets: position over contents. PageUp and PageDown page through windows; Esc closes them.

The status tables are stricter about fog than the map is: a foreign **world** is listed only if scouted, since the tables are made of numbers and knowing a world exists tells you none of them. A merely *known* enemy **fleet** does get a row, reading (unknown) and (out of range) – being warned that something is out there is the point of that window.

8.2 Direct keys on the map

Key	Action
arrows / <code>h j k l</code>	Move the cursor
<code>f</code>	Fleet roster
<code>b</code>	Construction screen
<code>w</code>	Warp link frequencies
<code>d</code>	Defense grid for the cursor’s world
<code>z</code>	Close-up of the cursor’s object
<code>i</code>	Production report for it
<code>x</code>	Self-destruct it (if it is yours, and you are certain)
<code>g</code>	The prologue menu
<code>m</code> or <code>F2</code>	The command menu bar
<code>n</code>	Next turn – end the year
<code>q</code>	Quit (asking first if unsaved)

The close-up and production reports always act on whatever the cursor is over, which is how the original reached them too.

8.3 The menu bar

Seven pull-downs. The letter in parentheses is the accelerator, counted exactly as the original wrote it – which is why `caNcel orders` answers to N and `auTo attack` to T.

Info (I)

- (A)bout Re:creon
- (D)OS shell

Game (G)

- (P)ause game
- (S)tatus hardcopy
- (N)ext turn
- (Q)uit

Empire (E)

- (S)end message
- (R)ead messages
- (T)rade technology
- (L)ink frequencies

Worlds (W)

- (C)lose up
- (D)esignate
- (P)roduction
- (I)SSP
- (N)ame
- (R)emove name
- (L)iberate
- (S)elf-destruct
- (T)erraform

Fleet (F)

- (D)eploy
- (C)hange destination
- (T)ransfer
- (A)bort/join
- (R)efuel
- (S)RM sweep
- (O)rders
- ca(N)cel orders
- (P)robe

Build (B)

- (S)ite status
- (N)ew construction site
- (A)bort construction

Ministry of War (M)

- (A)ttack
- au(T)o attack
- (L)aunch LAMs
- (D)efenses

Status hardcopy is the game's old printout – *Game ▶ Status hardcopy* builds the full report of your worlds, fleets and holdings, and lets you write it to a file. Its numbers are encoded the way a dot-matrix printer had room for: your own holdings in hundreds (so `42` means about 4200, and `++` is past 99), while another empire's read as estimates – `no`, `y` plus thousands, or `y+`, with raw materials as `--` because you cannot see them at all.

8.4 Interactive combat keys

The grouping screen, first. Build up to nine groups out of the ships and cargo in the attacking fleet:

Key	Action
<code>up</code> / <code>down</code>	Select group
<code>left</code> / <code>right</code>	Change its ship type
<code>space</code>	Auto-fill cargo
<code>enter</code>	Load an amount
<code>m</code> / <code>n</code>	Load men / ninjas
<code>Esc</code>	Confirm the grouping and fight

Then, round by round:

Key	Action
<code>e</code>	Engage – fight the round
<code>m</code>	Manoeuvre a group: advance, retreat or hold
<code>t</code>	Assign a group's target
<code>d</code>	Damage report from last round
<code>r</code>	Retreat the attack

Groups begin in deep space aiming at nothing. A player who only presses `e` fights a battle in which nobody closes and nobody fires, forever: `m` and `t` are the decisions the screen exists for.

8.5 Construction and defense screens

`b` lists the world's sites: `c` starts a new construction (pick the kind, pay the materials, point at the object), `a` aborts one. `w` opens the warp-link frequency screen, where each gate and link of yours is retuned – both ends have to agree, and retuning can strand your own fleets mid-network.

`d` opens the defense grid: four fixed types, five shells, each type's five percentages forced to total 100. The normalizer's fallback layer is **sub-orbit**: ground percentages for things that cannot land, any shortfall, and any rounding remainder all park there. Percentages typed outside 0..100 store as zero, not as the nearest bound.

8.6 Fleet roster keys

The roster (`f`) lists your scouted fleets. Selecting one and pressing enter opens its detail screen, where the Fleet menu items act on it directly; the transfer, abort and refuel screens all pick their counterpart fleets from the same sector.



9 Scenarios

Every scenario is an argument about history, written by someone who expects to be contradicted. – *attributed to the authors of the original scenarios, in jest*

A galaxy is not generated; it is *authored*. Every game of Re:creon starts from a scenario file – a plain-text list of worlds, owners, fleets, nebulae, minefields, empires and prose, loaded in one forward pass that asks you for the player count and your empire's name *while* it reads. There is no procedural “random map” button, because the original never had one.

9.1 The fourteen you ship with

Thirteen adapt scenarios from Anacreon: Reconstruction 4021. Each has a new title and a revised introduction. *Corsairs of Meridian* and *Vhalsecc* also rename places and empires. *The Long Sleep* and *The Four Heirs* have small repairs so they can load. The repository's `src/recreon/data/scenarios/README.md` records the changes. The fourteenth, *frontier.scn*, is new content written for this recreation.

File	Title	Original name
<code>frontier.scn</code>	The Kalgan Frontier	– (authored for Re:creon)
<code>firstlight.scn</code>	First Light: 4021	Reconstruction: 4021
<code>twocrowns.scn</code>	Two Crowns	East vs. West
<code>triad.scn</code>	Triad	Trinity
<code>fourheirs.scn</code>	The Four Heirs	The Four Princes
<code>outerreach.scn</code>	The Outer Reach	The Periphery
<code>corsairs.scn</code>	Corsairs of Meridian	The Pirates of Jakarta
<code>boundary.scn</code>	The Boundary Stones	Algerian Fences
<code>reckoning.scn</code>	The Reckoning	The Aftermath
<code>longrun.scn</code>	The Long Run	The Gauntlet
<code>longsleep.scn</code>	The Long Sleep	A Rude Awakening
<code>thousandsuns.scn</code>	The Thousand Suns	Imperium Galactica
<code>veil.scn</code>	The Veil	The Nebula
<code>vhalsecc.scn</code>	Vhalsecc	Arronax

First game? *First Light* is the introduction the original boxed with the manual: a small, legible galaxy, and the scenario the manual's advice was written against. *Corsairs of Meridian* is where to go once you want the pirates to teach you about convoy routes; *The Veil* is the nebula-warfare exam, authored around a galaxy of dust you cannot scout through.

9.2 What the introductions are, and where the prose comes from

Every scenario opens with one or more pages of background text – read them; in several cases they contain hints that are otherwise only discoverable by losing something. The bundled introductions were revised from the recovered originals.

The same files also carry **per-object background**: seven of the shipped galaxies index prose blocks to specific worlds, starbases and stargates, so closing up on a named place (`z`) or conquering it can print the author's words for *that object* instead of a generic report. Some references in the *Corsairs of Meridian* and

Vhalsecc background passages were changed with their place names. When you take a place that has a passage written about it, the conquest screen says so.

9.3 Scenario load failures

The original *AWAKEN* and *PRINCES* files have defects that prevent them from loading. Their bundled counterparts, *The Long Sleep* and *The Four Heirs*, include the two repairs described in the scenario provenance file. A separate placement failure can still occur on an unlucky random roll:

Each scenario places a mix of hand-authored and randomly-rolled worlds, and the random rolls ask for open sectors inside their zone – up to a hundred attempts, and then they give up rather than stack two worlds in one sector. The densest galaxies (*The Long Run*, *The Long Sleep*, packed at 200 worlds) will therefore refuse some rolls: roughly one in fifteen in the denser cases. This is the bundled scenario density and the original's placement limit – the DOS build could fail on the same rolls. **Start a new game; it will place.**

9.4 How a new game is assembled

1. Choose the scenario. The header (its declared format, its title) is checked before anything else happens; files from the original declare version 10.
2. The introduction pages scroll by.
3. Choose how many empires are run by human players – one, two, or more – and which slots those humans take.
4. Name each human empire, or take a random one from the original's table of 59. Computer empires are seated as their scenario specifies and pick up personalities at load: pirate, kingdom, guardian or berserker, each with its own temperament for escalating from raiding to annihilation.

Randomness in world placement, combat and events all runs through a ported copy of the original's own generator, seeded by the clock (the original did the same – the galaxies players saw in 2004 never existed twice), so a scenario can genuinely unfold differently on every run.

9.5 Writing your own

The shipped files are plain text and readable: directives for zones, `CreateWorld` blocks, empire seats, nebulae, the `BEGINTEXT` / `ENDTEXT` prose blocks. Copy the smallest one that looks like the game you want and start editing – they are lightly commented, and the scenario parser's test suite is a catalogue of every accepted directive. A scenario that disagrees with the format loses, every time – the original files are the specification.



10 Glossary

The empire names a thing twice: once for the record, once for the crew that has to carry it. –
Concordance of the Reconstruction, preface

Sector

One cell of the square map grid – a star system and whatever sits in it: worlds, bases, gates, fleets, phenomena. The unit of all distances.

Chebyshev distance

The map's measure: the larger of the horizontal and vertical gaps between two sectors, so diagonal movement costs exactly the same as straight movement.

Capital

An empire's seat world. All your typed coordinates are measured from it, and it counts as a laboratory in every research roll.

Designation

The assigned job of a world – agricultural, mine, shipyard base, university, outpost, and so on. Sets its principal industry and how its leftover industry is distributed.

Class

What a world physically is: Paradise, Ice world, Gas Giant, Ancient ruins. Sets trillum reserves, ground-defense bonus, and per-industry efficiency. Changed only by terraforming.

Scouted

The strongest grade of knowledge: you have seen the object and can see what it holds. Merely **known** objects are on the map but opaque; the detail panels and status tables demand scouting.

Base scan

The slow per-object coin flip a starbase runs each year, and the only way to discover something you had no idea existed.

Probe

A single-use sensor packet (*Fleet ► Probe*) sent to a known destination; reports what is there and returns by next year.

ISSP

Import/export self-sufficiency settings – four sliders per world (chemicals, metal, supplies, trillum), 0 to 10, where 5 means the world produces exactly what it consumes.

Supply link

The habit of food factories, mines and complexes of trading raw material across the eight neighbouring sectors. Starvation is usually a link problem before it is a fleet problem.

Trillum

The only fuel, and the rarest raw material. Ships, bases and jump drives burn it; worlds hold finite reserves set by their class. A fleet reporting *out of trillum* is a monument.

Fleet type

Standard, advanced-warp, penetrator, jump or hunter-killer force – decided by what the fleet lacks, not what it has, and setting the fleet's sectors-per-year movement rate.

Jump

Movement through the gate network or a jump base's projection. Gate jumps cover any distance in one year. Jump traffic trips minefields; warp traffic does not.

Link (warp link)

A paired corridor structure, five years to build: a gate whose network is exactly one other end.

Disrupter

A structure that denies enemy jump movement in its neighbourhood while letting friendly warp fleets ride at ten sectors a year.

Group

Up to nine formations per attack: a set number of one ship type with its loaded troops. Groups manoeuvre and aim independently; this is the unit you command on the combat screen.

Shell

One of five altitudes – deep space, high orbit, orbit, sub-orbit, ground. Combat resolves outermost first; anything not placed deliberately ends up parked in sub-orbit.

SRM field

Space-minefield; a site-built hazard that damages fleets making jump moves through its sector, and can be swept.

Command base

The mobile starbase. Scans, scouts, refuels, supports jump traffic, and shuffles one sector a year under orders.

Fortress

The base that multiplies sector defense (combat adjustment 250%) and catapults departing fleets four sectors forward.

Industrial complex

A base that concentrates the surrounding sectors' raw material and passes its own overflow back to the mines.

GDM

Ground-defense missile. A world's fixed salvo, sized by technology, fired at anything in range of the surface.

LAM

Light attack missile. Precision first-strike stock carried by bases, fired at enemy fleets or a world's defenses; not interceptable, not recoverable.

Ninja

A legion that costs metal to produce, counts five-to-one against revolution and occupation duties, and wins fights they should lose.

Ambrosia

The product of bio-tech labs at 11.5 kilotons per billion people. Withholds: death and rage. Delivers: a 45% industry bonus from an addicted, satisfied population.

Revolution index

Per-world unrest, 0..100. Above 75, rebellion is possible; garrisons cap it, and quiet worlds under occupation resent the garrison.

Rebellion

The uprising that follows: either crushed by local troops, or successful, and the world becomes independent.

Auto-attack

The computer's attack routine – the tactical AI that closes, aims and fires your fleets for you when you choose *auTo attack*, and that attacks you when it thinks the numbers are good.

Personality (NPE empire)

The computer empires' temperaments – pirate, kingdom, guardian, berserker – set by the scenario and reflected in what they build, where they raid, and how fast they escalate.

Ruins world

A relic of the old empire: a small research bonus, an easy conquest, and scenarios that disagree about what else.

Wanderer

An unidentified moving contact in a sensor report. The original reserved the name; you may go a hundred games without meeting one that exists.



11 Reference Tables

Every table in this appendix is taken directly from the game's own data – the same constants the simulation uses, transcribed from *DATA CNST.PAS* and *ATTACK.PAS*. They are reproduced here so that a number you read in the manual and a number the game computes are never two different numbers.

Units follow the original's conventions:

- **Population** is counted in units of 100 million.
- **Fuel figures** for ships are scaled: consumption is recorded in thousandths of a unit per sector per ship, capacity in hundredths of a unit per ship.
- **Raw-material costs** are quantities (megatons, except ambrosia and trillum in kilotons) needed to build **100 units** of the thing.
- **Percentages of population in the military** are stored in hundredths of one percent.
- A dash means the game has no entry for that cell; a zero is a real zero.

11.1 Technology Levels

There are eleven technology levels. **Base population** is the average population a world of that technology supports at 50% efficiency, in units of 100 million. **TIP adj** and **prod adj** are the percent multipliers the level applies to a world's industrial potential and production; **min supply ind** is the percent of industry a world needs in food factories to be self-sufficient; **military pct** is the percent of population in the military at military index 50 (hundredths of a percent). **GDM salvo** is how many ground-defense missiles a world can put up at once. The **Unlocks** column lists the things first buildable at that level.

Lvl	Level	Base pop	TIP adj	Prod adj	Min sup ind	Military pct	GDM salvo	Unlocks
0	pre-tech	3	25	12	95	1	0	supply
1	primitive	10	40	24	90	80	0	troop, metal
2	pre-atomic	100	49	36	80	100	0	chemical
3	atomic	250	57	47	70	90	20	GDM, trillum
4	pre-warp	500	66	58	50	80	217	fighter
5	warp	700	80	67	40	75	662	transport
6	jump	1100	85	76	34	60	1013	ion cannon, jumpship, jumtransport
7	bio-tech	1700	90	84	30	50	1261	defense satellite, hunter- killer, penetrator, ambrosia, outpost
8	starship	2000	94	90	28	30	2163	LAM, starship, ninja, SRM, command base, industrial complex
9	pre-gate	2500	97	95	26	15	2644	fortress, link, disrupter
10	gate	3000	100	100	23	10	2759	gate, terraforming

11.2 Industries and Output

Nine kinds of industry, the technology needed to run them, and the metal cost of building 100 points of each.

Industry	Min tech	Metal per 100
bio-tech labs	bio-tech	100
chemical plants	pre-atomic	500
metal mines	primitive	100
ship yards	pre-warp	1900
jumpship yards	jump	1200
starship yards	starship	1500
transport yards	warp	1000
food factories	pre-tech	0
trillum mines	atomic	300

What each industry produces, in relative points per 100 points of that industry. Bio-tech labs produce ambrosia; the food factories produce supplies; the raw-material industries their eponymous materials; the

yards build hulls. Heads: men and nnj are legions, amb and tri kilotons, che, met and sup megatons; hull abbreviations are fgt, hkr, jmp, jtn, pen, ssp, trn.

Industry	fgt	hkr	jmp	jtn	pen	ssp	trn	men	nnj	amb	che	met	sup	tri
bio-tech labs	—	—	—	—	—	—	—	—	10	175	—	—	—	—
chemical plants	—	—	—	—	—	—	—	—	—	—	175	—	—	—
metal mines	—	—	—	—	—	—	—	—	—	—	—	350	—	—
ship yards	27	5	9	5	4	2	10	—	—	—	—	—	—	—
jumpship yards	—	25	50	30	—	—	—	—	—	—	—	—	—	—
starship yards	—	—	—	—	40	15	—	—	—	—	—	—	—	—
transport yards	75	—	—	—	—	—	45	—	—	—	—	—	—	—
food factories	—	—	—	—	—	—	—	—	—	—	—	—	320	—
trillum mines	—	—	—	—	—	—	—	—	—	—	—	—	—	75

11.3 Ships

Combat power is 0..100 on an absolute scale (a starship anchors the top end at 100). **Weapon eff** is the average effectiveness of the type's weapons; **value** and **protection** are the AI's bookkeeping figures – how many points a ship of the type is worth and how much protection it offers or needs. **Trns** is cargo capacity relative to a transport. Fuel figures are scaled as noted above.

Ship	Tech	Power	Weap eff	Value	Prot off	Prot need	Trns	Fuel/sector	Fuel cap
fighter	pre-warp	1	30	5	5	5	0	10	3
hunter-killer	bio-tech	20	100	50	10	30	0	1086	851
jumpship	jump	12	100	50	10	20	0	659	329
jumptransport	jump	1	10	30	1	50	0.2	894	1341
penetrator	bio-tech	25	150	100	50	200	0	943	1886
starship	starship	100	100	200	100	500	0	1427	2854
transport	warp	0	10	30	0	150	1	994	1988

11.4 Cargo and Defenses

Cargo carried in the hold of a transport. **Space per transport** is how many units of that cargo one standard transport carries; a jumptransport carries a fifth of it. **Power** is on the surrender scale.

Cargo	Power	Space per transport	Fuel/sector
troop	20	5	10
ninja	100	5	10
ambrosia	—	100	15
chemical	—	3	10
metal	—	3	10
supply	—	2	15
trillum	—	100	20

The four things a world can build as fixed defense.

Defense	Tech	Power	Weap eff	Def adj	Build rate	che / met / tri
LAM	starship	100	100	145	220	25 / 20 / 5
defense satellite	bio-tech	100	150	76	50	30 / 140 / 20
GDM	atomic	10	100	215	250	10 / 20 / 2
ion cannon	jump	50	30	83	95	25 / 150 / 10

11.5 Construction

Years to complete, and the cargo (megatons; ambrosia and trillum kilotons) a site consumes **per year** while under construction.

Structure	Years	troop	ninja	ambrosia	chemical	metal	supply	trillum
SRM	2	—	—	—	110	500	—	80
command base	6	—	—	—	460	2300	—	180
fortress	12	—	—	—	840	2870	—	250
industrial complex	10	—	—	—	590	2600	—	150
outpost	3	—	—	—	350	1120	—	150
gate	15	—	—	—	2530	3920	—	1450
link	5	—	—	—	1560	2550	—	290
disrupter	8	—	—	—	1110	1180	—	1120

11.6 Raw-Material Costs

Raw materials consumed to build 100 units of each thing. (Structures are built by year and are in the construction table instead; their per-year rate scales the same way.)

Thing	che	met	sup	tri
ambrosia	110	0	0	0
fighter	5	30	0	2
hunter-killer	100	110	0	18
jumpship	65	70	0	12
jumprtransport	100	110	0	16
penetrator	95	275	0	20
starship	175	520	0	30
transport	90	600	0	10
troop	0	0	0	0
ninja	50	0	0	0
LAM	25	20	0	5
defense satellite	30	140	0	20
GDM	10	20	0	2
ion cannon	25	150	0	10

11.7 Combat

11.7.1 The combat table

Destructive power of attacker against defender. At the standard adjustments, 100 points of accumulated damage destroys one unit, 200 destroys two, and so on.

Attacker defender	None	Lam	Def	Gdm	Ion	Fgt	Hkr	Jmp	Jtn	Pen	Ssp	Trn	Men	Nnj
—	0	0	0	0	0	0	0	0	0	0	0	0	0	0
LAM	0	200	100	200	150	250	60	75	100	50	15	200	0	0
defense satellite	0	0	0	0	0	25	90	100	100	50	6	250	0	0
GDM	0	0	0	0	0	250	100	100	100	50	25	50	0	0
ion cannon	0	0	0	0	0	50	65	75	75	15	2	100	0	0
fighter	0	5	5	1	25	25	15	8	20	12	1	25	5	4
hunter-killer	0	12	40	38	20	50	50	75	50	50	4	75	0	0
jumpship	0	8	15	25	10	150	25	38	50	25	1	150	0	0
jumptransport	0	0	1	0	0	25	1	1	5	0	0	10	0	0
penetrator	0	15	35	50	35	50	65	105	125	50	5	250	0	0
starship	0	45	90	200	50	1000	250	300	500	100	15	500	0	0
transport	0	0	0	0	0	12	0	0	1	0	0	5	0	0
troop	0	0	0	0	0	5	0	0	0	0	0	0	7	2
ninja	0	0	0	0	0	8	0	0	0	0	0	0	20	10

11.7.2 Military power (surrender test)

Powers used when weighing whether a beaten world surrenders.

Weapon	Power
LAM	80
defense satellite	75
GDM	20
ion cannon	65
fighter	2
hunter-killer	15
jumpship	10
jumptransport	1
penetrator	20
starship	100
transport	1
troop	20
ninja	100

11.7.3 Technology adjustment

Percent applied to the attacker's power for attacker technology (rows) against defender technology (columns). 100 is no adjustment.

Attacker defender	pre- tech	primitive	pre- atomic	atomic	pre- warp	warp	jump	bio- tech	starship	pre- gate	gate
pre- tech	100	90	50	25	10	5	3	1	1	1	1
primitive	120	100	80	40	25	10	5	3	1	1	1
pre- atomic	150	110	100	60	40	25	10	5	3	1	1
atomic	160	140	130	100	60	40	30	20	10	5	3
pre- warp	200	170	145	120	100	90	85	75	70	55	45
warp	250	200	175	140	110	100	95	90	85	70	55
jump	280	210	190	175	115	110	100	97	93	80	70
bio- tech	320	250	210	195	120	115	105	100	95	90	80
starship	360	320	280	220	125	120	115	110	100	97	90
pre- gate	500	400	300	250	130	125	120	115	110	100	95
gate	530	415	310	260	135	130	125	120	115	110	100

11.7.4 GDM interception

Ground-defense missiles shot down on approach per 10 attacking ships of the type.

Ship	GDMs killed
fighter	0
hunter-killer	3
jumpship	2
jumptransport	1
penetrator	15
starship	20
transport	0

11.8 World Classes

The world's class sets its trillum reserve, how strongly its terrain aids ground defense, and the percent of each industry that is effective on it. Industry column heads: Bio (labs), Che (micals), Min (es), SYG / SYJ / SYS / SYT (the four ship yards), Sup (plies), Tri (llium).

Class	Glyph	Min tech	Tri res.	Gd def.	Bio	Che	Min	SYG	SYJ	SYS	SYT	Sup	Tri
Ambrosia	A	pre-tech	200	100	100	100	75	100	100	100	100	100	75
Arid	a	primitive	350	100	100	80	100	100	100	100	100	85	100
Artificial	0	jump	50	100	100	40	40	250	200	300	300	40	40
Barren	B	pre-warp	1200	50	100	60	175	100	100	100	100	40	175
Class j	j	pre-tech	700	100	100	120	120	100	100	100	100	100	90
Class k	k	pre-tech	800	100	100	90	120	100	100	100	100	100	100
Class l	l	pre-tech	900	100	100	100	100	100	100	100	100	90	120
Class m	m	pre-tech	800	100	100	100	100	100	100	100	100	120	90
Desert	D	primitive	1500	140	100	60	80	100	100	100	100	60	190
Earth-like	E	pre-tech	800	100	100	100	100	100	100	100	100	100	100
Forest world	F	pre-tech	500	140	100	120	100	100	100	100	100	145	100
Gas Giant	G	pre-warp	450	50	100	150	50	150	125	175	150	40	60
Hostile life	h	pre-atomic	500	100	100	100	100	100	100	100	100	100	100
Ice world	I	pre-atomic	900	160	100	90	80	100	100	100	100	60	80
Jungle world	J	pre-tech	600	150	100	130	100	100	100	100	100	125	100
Ocean world	O	pre-warp	300	80	100	135	40	100	100	100	100	130	40
Paradise	1	pre-tech	600	100	120	150	125	100	100	100	100	200	150
Poisonous	P	atomic	350	60	100	200	80	100	100	100	100	40	80
Ancient ruins	2	pre-tech	200	100	100	100	100	100	100	100	100	100	100
Underground	U	primitive	900	130	100	100	150	100	100	100	100	70	125
Terraforming	X	warp	800	100	1	1	1	1	1	1	1	10	1
Volcanic	V	pre-atomic	1100	120	100	125	175	100	100	100	100	75	150

11.9 World Designations

The technology needed before a world may be designated, its principal industry, and the optimum military index for that designation (hundredths of a percent of population in uniform).

Designation	Glyph	Min tech	Principal industry	Opt military
agricultural world	a	pre-tech	food factories	5
ambrosia world	A	bio-tech	bio-tech labs	100
base planet	b	pre-warp	ship yards	200
base planet	B	gate	ship yards	200
capital	C	jump	ship yards	200
chemical planet	c	pre-atomic	chemical plants	10
independent world	i	pre-tech	ship yards	80
jumpship base	j	jump	jumpship yards	100
jumpship base	J	gate	jumpship yards	100
metal mine	m	primitive	metal mines	20
ninja world	N	starship	bio-tech labs	150
outpost	o	gate	ship yards	0
raw material mine	r	primitive	metal mines	20
raw material mine	R	gate	metal mines	20
starship base	s	bio-tech	starship yards	150
starship base	S	gate	starship yards	150
transport base	t	pre-warp	transport yards	80
transport base	T	gate	transport yards	80
university world	U	jump	metal mines	1
terraforming	X	warp	metal mines	10
trillum mine	z	atomic	trillum mines	30

11.10 Fleets and Movement

Fleet movement rate in sectors per year, by fleet type. A fleet's type is the slowest ship it contains.

Fleet type	Sectors / year
Standard fleet	1
Jump fleet	10
Hunter-killer force	10
Penetrator force	2
Advanced-warp fleet	2

Other movement and cargo constants:

- Jump and jump-transport bases project a fleet of up to **5 transports** worth of cargo (or the ship-count equivalent).
- Fuel per ton of cargo: **265** (thousandths).
- A standing fleet in space burns its type's fuel consumption divided by **10** per turn.
- Warp distance is Chebyshev: a diagonal sector costs the same as an orthogonal one.

11.11 Game Limits

Thing	Maximum
Worlds	200
Starbases	100
Fleets, total	240
Fleets per empire	30
Stargates	50
Construction sites	50
Probe packets per empire	10
Names per empire	100
Groups per attack	9
Max resources	9999